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B.Tech CSE, VIT Vellore · Software, systems & applied AI

EDUCATION

Vellore Institute of Technology (VIT), Vellore · B.Tech, Computer Science and Engineering — 2023 – 2027
CGPA 7.8 / 10 · Coursework: Data Structures & Algorithms, Operating Systems, Compiler Design, Computer Architecture, Parallel Computing

FAIPS Delhi Public School, Kuwait · Class XII: 87% · Class X: 95% — 2023

TECHNICAL SKILLS

Languages	Python, Java, C, C++, JavaScript (ES6+), TypeScript, SQL, Verilog, HTML5, CSS3
Backend & Cloud	REST APIs, Node.js, FastAPI, Flask, Spring Boot, Cloudflare Workers, D1, R2, Vercel, Supabase, Firebase, CI/CD
Frontend	React, Vite, Tailwind CSS, IndexedDB, offline-first sync
AI / ML	Transfer learning (MobileNetV3), Random Forest, LLM APIs (Gemini, Groq), RAG, LLM-as-judge evaluation, Ollama, Qiskit
Systems & Data	Streaming algorithms (Count-Min Sketch), CPU-GPU scheduling, FPGA, MySQL, SQLite, Linux, Git
Security	OWASP Top 10, OWASP ASVS, OWASP LLM Top 10, prompt-injection testing, post-quantum cryptography

PROJECTS

BidirFlow · Real-time wash-trading detection — Oct 2026
`Python · Verilog · FPGA`

- Built a streaming detector that flags wash trading, layering and circular trading rings on live Ethereum transfers in constant **32 MiB** memory at **213,000 transfers/s** (10,800× real time); patent application in preparation.
- Achieved **0.963 average precision** on 849,118 real transfers ($61B) vs. 0.009 for a Count-Min volume baseline; held up on held-out and pre-registered windows (0.90, 0.94) and on Polygon (0.95).
- Implemented the sketch core in Verilog, verified it bit for bit against the Python model, and placed and routed it on a Lattice ECP5 FPGA at **77.6 MHz**.
- Shipped a live dashboard answering natural-language queries via a local LLM (Ollama) that calls detector tools.

FieldProof · Offline-first field-evidence platform — Oct 2026
`React · TypeScript · Cloudflare Workers · D1 · R2`

- Deployed a fieldwork app (cases, visits, uploads, role-based review) on a serverless edge backend with server-side RBAC.
- Designed offline IndexedDB drafts with duplicate-safe sync (operation IDs, version checks) and a cited Gemini assistant.
- Ran an OWASP Top 10 / ASVS L1 review of 20 API routes (Semgrep, gitleaks); fixed 10 of 15 findings, cutting critical advisories from 1 to 0; verified by **90 automated tests** in CI.

PromptShield · Prompt-injection tester for LLM chatbots — Oct 2026
`React · Vercel · Groq · Supabase`

- Built solo for LovHack Season 3: runs **25 attacks in 5 categories**, scores leaks with canary tokens and an LLM judge, then rewrites defenses and reruns the suite.

RESEARCH & PUBLICATIONS

Q-SOC · Quantum-aware SOC · co-author, IEEE paper under submission — 2026
`Python · FastAPI · Qiskit`

- Co-built a SOC with a rule engine (17 rules, MITRE ATT&CK tags), attack simulator, Quantum Readiness Index, Qiskit Grover/Shor demos and ML-DSA-65 post-quantum report signing.
- Validated detection with **220/220** passing runs across 11 scenarios and a 33/33 negative control; raised ingestion from 35 to **1,526 events/s** by batching.

RescueSched · CPU-GPU inference scheduling · first author — 2026 – 2027
`Python · Flask · MobileNetV3`

- Designed a disaster-image scheduler that re-prioritizes queued requests by hazard severity from a fine-tuned MobileNetV3.
- Predicts runtimes with a random forest and picks CPU or GPU and batch size (1 to 64) by a deadline-weighted cost.

CERTIFICATIONS & ACHIEVEMENTS

- Oracle OCI Certified AI Foundations Associate; OCI Certified Foundations Associate; GitHub Foundations (2026)
- Booking Holdings HackCellence 2026: shortlisted for Round 2 of a national hiring hackathon; hackathon team lead since 2024
- Football captain: interschool title (Class XII), college tournament win; House Captain, DPS Kuwait